Online Appendix: Nonconvex Optimization with Spectral Radius Regularization

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Appendix A
Hessian-Vector Operations Derivation

The forward computation for each layer of a network with input x, output y, weights w, activation σ, bias I, error or loss measure $E = E(y)$, and direct derivative $e_k = dE/dy_k$ is given by:

$$x_k = x_k(y_{k-1}, w_k) = \sum_j w_{jk} y_{j(k-1)}$$
$$y_k = y_k(x_k, I_k) = \sigma_k(x_k) + I_k$$

The backward computation:

$$\frac{\partial E}{\partial y_k} = e_k(y_k) + \sum_j w_{jk} \frac{\partial E}{\partial x_j}$$
$$\frac{\partial E}{\partial x_k} = \sigma_k'(x_k) \frac{\partial E}{\partial y_k}$$
$$\frac{\partial E}{\partial w_{jk}} = y_k \frac{\partial E}{\partial x_j}$$

Applying $\mathcal{R}_v\left\{\cdot\right\}$ to forward pass:

$$\mathcal{R}_v\left\{x_k; w\right\} = \sum_j \left(w_{jk}\mathcal{R}_v\left\{y_{j(k-1)}; w\right\} + v_{jk} y_{j(k-1)}\right)$$
$$\mathcal{R}_v\left\{y_k; w\right\} = \mathcal{R}_v\left\{x_k; w\right\}\sigma_k'(x_k)$$

The backward computation follows as:

$$\begin{aligned}
\mathcal{R}_v\left\{\frac{\partial E}{\partial y_k}; w\right\} =& e_k'(y_k)\mathcal{R}_v\left\{y_k; w\right\} \\
&+ \sum_j \left[w_{jk}\mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\} + v_{jk}\frac{\partial E}{\partial x_j}\right] \\
\mathcal{R}_v\left\{\frac{\partial E}{\partial x_k}; w\right\} =& \sigma_k'(x_k)\mathcal{R}_v\left\{\frac{\partial E}{\partial y_k}; w\right\} \\
&+ \mathcal{R}_v\left\{x_k; w\right\}\sigma_k''(x_k)\frac{\partial E}{\partial y_k} \\
\mathcal{R}_v\left\{\frac{\partial E}{\partial w_{jk}}; w\right\} =& y_k\mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\} + \mathcal{R}_v\left\{y_k; w\right\}\frac{\partial E}{\partial x_j}
\end{aligned}$$

This yields the result found in [1]. However, we extend it one step further by applying $\mathcal{R}_v\left\{\cdot\right\}$ again, i.e., applying $\mathcal{R}_v^2\left\{\cdot\right\} = \mathcal{R}_v\left\{\mathcal{R}_v\left\{\cdot\right\}\right\}$ to the original forward pass:

$$\begin{aligned}
\mathcal{R}_v^2\left\{x_k; w\right\} =& \sum_j \Big[w_{ji}\mathcal{R}_v^2\left\{y_{j(k-1)}; w\right\} \\
&+ 2v_{jk}\mathcal{R}_v\left\{y_{j(k-1)}; w\right\}\Big] \\
\mathcal{R}_v^2\left\{y_k; w\right\} =& \mathcal{R}_v^2\left\{x_k; w\right\}\sigma_k'(x_k) \\
&+ \left(\mathcal{R}_v\left\{x_k; w\right\}\right)^2\sigma_k''(x_k)
\end{aligned}$$

The backward computation follows as:

$$\begin{aligned}
\mathcal{R}_v^2\left\{\frac{\partial E}{\partial y_k}; w\right\} =& e_k''(y_k)\left(\mathcal{R}_v\left\{y_k; w\right\}\right)^2 \\
&+ e_k'(y_k)\mathcal{R}_v^2\left\{y_k; w\right\} \\
&+ \sum_j \left[w_{jk}\mathcal{R}_v^2\left\{\frac{\partial E}{\partial x_j}; w\right\} + \right. \\
&\left. 2v_{jk}\mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\}\right] \\
\mathcal{R}_v^2\left\{\frac{\partial E}{\partial x_k}; w\right\} =& 2\mathcal{R}_v\left\{x_k; w\right\}\sigma_k''(x_k)\mathcal{R}_v\left\{\frac{\partial E}{\partial y_k}; w\right\} \\
&+ \sigma_k'(x_k)\mathcal{R}_v^2\left\{\frac{\partial E}{\partial y_k}; w\right\} \\
&+ \mathcal{R}_v^2\left\{x_k; w\right\}\sigma_k''(x_k)\frac{\partial E}{\partial y_k} \\
&+ \left(\mathcal{R}_v\left\{x_k; w\right\}\right)^2\sigma_k'''(x_k)\frac{\partial E}{\partial y_k} \\
\mathcal{R}_v^2\left\{\frac{\partial E}{\partial w_{jk}}; w\right\} =& 2\mathcal{R}_v\left\{y_k; w\right\}\mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\} \\
&+ y_k\mathcal{R}_v^2\left\{\frac{\partial E}{\partial x_j}; w\right\} \\
&+ \mathcal{R}_v^2\left\{y_k; w\right\}\frac{\partial E}{\partial x_j}
\end{aligned}$$

The original formulation $\mathcal{R}_v\left\{\cdot\right\}$ allows us to efficiently compute $H(w)v$, which can be used to compute $\rho(w)$ and/or estimate the eigenvector $\bar{v}$ corresponding to the spectral radius (via power iteration or LOBPCG). However, the extended formulation $\mathcal{R}_v^2\left\{\cdot\right\}$ allows us to efficiently compute $v^T\nabla H(w)v$ and thus $\nabla\rho(w)$. This enables us to efficiently compute the gradient of our optimization problem for use in gradient descent methods.

Appendix B
Convergence Analysis Proofs

A. Stochastic Gradient Descent Convergence

First, we prove the convergence of our regularization term (Lemma 4.2).

Proof. We start by splitting $v_k^T \nabla H_k v_k$ into its components

$$\begin{aligned} v_k^T \nabla H_k v_k =& (v_k - \bar{v}_k + \bar{v}_k)^T \nabla H_k (v_k - \bar{v}_k + \bar{v}_k) \\ =& (v_k - \bar{v}_k)^T \nabla H_k (v_k - \bar{v}_k) \\ &+ 2(v_k - \bar{v}_k)^T \nabla H_k \bar{v}_k + \bar{v}_k^T \nabla H_k \bar{v}_k. \end{aligned}$$

The last term $\bar{v}_k^T \nabla H_k \bar{v}_k = \nabla \bar{\rho}_k$ by definition. We apply the triangle inequality and bind the other terms. Given the convergence criteria on v_k and Assumptions A1 and A4 (with $||H(w) - H(\omega)|| \leq L||w - \omega||,\ \forall\ w, \omega \in \mathbb{R}^n, L \geq 0$), it follows that

$$\left|\left|(v_k - \bar{v}_k)^T \nabla H_k \bar{v}_k\right|\right| \leq L\,||v_k - \bar{v}_k|| \leq L\varepsilon_k.$$

For the first term, we similarly get

$$\begin{aligned} \left|\left|(v_k - \bar{v}_k)^T \nabla H_k (v_k - \bar{v}_k)\right|\right| \leq& L\,||v_k - \bar{v}_k||^2 \\ \leq& L\varepsilon_k^2. \end{aligned}$$

Given Assumption A5, the limit

$$\lim_{k\to\infty} v_k^T \nabla H_k v_k = \lim_{k\to\infty} \bar{v}_k^T \nabla H_k \bar{v}_k = \lim_{k\to\infty} \nabla \bar{\rho}_k.$$

□

Next, we prove that Algorithms 1 and 2 follow the assumed update steps (Lemma 4.3):

Proof. We begin by splitting p_k into its components

$$\begin{aligned} ||p_k||^2 =& ||p_k - \nabla g_k + \nabla g_k||^2 \\ =& ||p_k - \nabla g_k||^2 + ||\nabla g_k||^2 \\ &+ 2\left(p_k - \nabla g_k\right)^T \nabla g_k. \end{aligned}$$

Let us first assume $\rho_k > K$. By the definition of g_k and the triangle inequality,

$$||\nabla g_k||^2 \leq ||\nabla f_k||^2 + 2\mu||\nabla f_k||||\nabla \rho_k|| + \mu^2||\nabla \rho_k||^2.$$

Taking the expectation (with respect to batch B_k conditioned on the history $\mathcal{P}_k$) and applying the Cauchy-Schwarz inequality yields

$$\begin{aligned} \mathbb{E}_{B_k}\left[||p_k||^2\middle|\mathcal{P}_k\right] \leq& \mathbb{E}_{B_k}\left[||\nabla f_k||^2\middle|\mathcal{P}_k\right] \\ &+ 2\mu\left(\mathbb{E}_{B_k}\left[||\nabla f_k||^2\middle|\mathcal{P}_k\right]\right)^{\frac{1}{2}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^2\middle|\mathcal{P}_k\right]\right)^{\frac{1}{2}} \\ &+ \mu^2\,\mathbb{E}_{B_k}\left[||\nabla \rho_k||^2\middle|\mathcal{P}_k\right]. \end{aligned}$$

Applying Hölder's inequality with $||\nabla f_k||^2$, $||\nabla \rho_k||$, $p = 3/2$, and $q = 3$ yields

$$\begin{aligned} \mathbb{E}_{B_k}\left[||\nabla f_k||^2||\nabla \rho_k||\middle|\mathcal{P}_k\right] \leq& \left(\mathbb{E}_{B_k}\left[||\nabla f_k||^3\middle|\mathcal{P}_k\right]\right)^{\frac{2}{3}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^3\middle|\mathcal{P}_k\right]\right)^{\frac{1}{3}}. \end{aligned}$$

Similarly, using $||\nabla f_k||^3$, $||\nabla \rho_k||$, $p = 4/3$, and $q = 4$ yields

$$\begin{aligned} \mathbb{E}_{B_k}\left[||\nabla f_k||^3||\nabla \rho_k||\middle|\mathcal{P}_k\right] \leq& \left(\mathbb{E}_{B_k}\left[||\nabla f_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{3}{4}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{1}{4}}. \end{aligned}$$

Applying this, we obtain

$$\begin{aligned} \mathbb{E}_{B_k}\left[||p_k||^3\middle|\mathcal{P}_k\right] \leq& \mathbb{E}_{B_k}\left[||\nabla f_k||^3\middle|\mathcal{P}_k\right] \\ &+ 3\mu\left(\mathbb{E}_{B_k}\left[||\nabla f_k||^3\middle|\mathcal{P}_k\right]\right)^{\frac{2}{3}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^3\middle|\mathcal{P}_k\right]\right)^{\frac{1}{3}} \\ &+ 3\mu^2\left(\mathbb{E}_{B_k}\left[||\nabla f_k||^3\middle|\mathcal{P}_k\right]\right)^{\frac{1}{3}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^3\middle|\mathcal{P}_k\right]\right)^{\frac{2}{3}} \\ &+ \mu^3\,\mathbb{E}_{B_k}\left[||\nabla \rho_k||^3\middle|\mathcal{P}_k\right], \end{aligned}$$

$$\begin{aligned} \mathbb{E}_{B_k}\left[||p_k||^4\middle|\mathcal{P}_k\right] \leq& \mathbb{E}_{B_k}\left[||\nabla f_k||^4\middle|\mathcal{P}_k\right] \\ &+ 4\mu\left(\mathbb{E}_{B_k}\left[||\nabla f_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{3}{4}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{1}{4}} \\ &+ 6\mu^2\left(\mathbb{E}_{B_k}\left[||\nabla f_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{1}{2}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{1}{2}} \\ &+ 4\mu^3\left(\mathbb{E}_{B_k}\left[||\nabla f_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{1}{4}} \\ &\times \left(\mathbb{E}_{B_k}\left[||\nabla \rho_k||^4\middle|\mathcal{P}_k\right]\right)^{\frac{3}{4}} \\ &+ \mu^4\,\mathbb{E}_{B_k}\left[||\nabla \rho_k||^4\middle|\mathcal{P}_k\right]. \end{aligned}$$

Given Assumption A3, this implies that

$$\mathbb{E}_{B_k}\left[||p_k||^j\middle|\mathcal{P}_k\right] \leq \overline{A}_j + \overline{B}_j||w_k||^j,$$

for $j = 2, 3, 4$ and some positive constants $\overline{A}_j, \overline{B}_j$. Combining this with the above results shows that the second, third, and fourth moments of the update term are bounded, as required.

If $p_k \leq K$, then $p_k = \nabla f_k$, and the statement follows from Assumption A3. □

The rest of this proof uses our assumptions and lemmas and follows the proof of Bottou [2] that SGD converges. In the next step, we prove confinement (Lemma 4.4).

Proof. Let $\varphi(x) := \begin{cases} 0, & x < D, \\ (x - D)^2, & x \geq D, \end{cases}$ and $\psi_k := \varphi(||w_k||^2)$. This implies that

$$\varphi(y) - \varphi(x) \leq (y - x)\varphi'(x) + (y - x)^2,$$

for $y, x \in \mathbb{R}$. Note that this becomes an equality when $x, y > D$.

Applying this to $\psi_{k+1} - \psi_k$, we derive

$$\begin{aligned} \psi_{k+1} - \psi_k \leq& \left(-2\alpha_k w_k^T p_k + \alpha_k^2||p_k||^2\right)\psi'(||w_k||^2) \\ &+ 4\alpha_k^2\left(w_k^T p_k\right)^2 - 4\alpha_k^3 w_k^T p_k||p_k||^2 \\ &+ \alpha_k^4||p_k||^4. \end{aligned}$$

By the Cauchy-Schwartz inequality, we get

$$\begin{aligned}\psi_{k+1}-\psi_k \leq &-2\alpha_k w_k^T p_k \psi'(||w_k||^2)\\ &+\alpha_k^2||p_k||^2\psi'(||w_k||^2)\\ &+4\alpha_k^2||w_k||^2||p_k||^2+4\alpha_k^3||w_k||||p_k||^3\\ &+\alpha_k^4||p_k||^4.\end{aligned}$$

Taking the expectation, we have

$$\begin{aligned}\mathbb{E}_{B_k}\left[\psi_{k+1}-\psi_k\middle|\mathcal{P}_k\right] \leq &-2\alpha_k w_k^T \nabla g_k \psi'(||w_k||^2)\\ &+\alpha_k^2\,\mathbb{E}_{B_k}\left[||p_k||^2\middle|\mathcal{P}_k\right]\psi'(||w_k||^2)\\ &+4\alpha_k^2||w_k||^2\,\mathbb{E}_{B_k}\left[||p_k||^2\middle|\mathcal{P}_k\right]\\ &+4\alpha_k^3||w_k||\,\mathbb{E}_{B_k}\left[||p_k||^3\middle|\mathcal{P}_k\right]\\ &+\alpha_k^4\,\mathbb{E}_{B_k}\left[||p_k||^4\middle|\mathcal{P}_k\right].\end{aligned}$$

Given Assumption A2, for sufficiently large k, $\alpha_k^2 \geq \alpha_k^3 \geq \alpha_k^4$. Due to Lemma 4.3, there exist positive constants A_0, B_0 such that

$$\begin{aligned}\mathbb{E}_{B_k}\left[\psi_{k+1}-\psi_k\middle|\mathcal{P}_k\right] \leq &-2\alpha_k w_k^T \nabla g_k \psi'(||w_k||^2)\\ &+\alpha_k^2\left(A_0+B_0||w_k||^4\right),\end{aligned}$$

and thus, there exist positive constants A, B such that

$$\begin{aligned}\mathbb{E}_{B_k}\left[\psi_{k+1}-\psi_k\middle|\mathcal{P}_k\right] \leq &-2\alpha_k w_k^T \nabla g_k \psi'(||w_k||^2)\\ &+\alpha_k^2\left(A+B\psi_k\right).\end{aligned}$$

If $||w_k||^2 < D$, then $\psi'(||w_k||^2)=0$, and the first term on the right-hand side is zero. If $||w_k||^2 \geq D$, by Assumption A6, the first term of the right-hand side is negative. Therefore,

$$\mathbb{E}_{B_k}\left[\psi_{k+1}-\psi_k\middle|\mathcal{P}_k\right] \leq \alpha_k^2\left(A+B\psi_k\right).$$

We then transform the expectation inequality to

$$\mathbb{E}_{B_k}\left[\psi_{k+1}-(1+\alpha_k^2 B)\psi_k\middle|\mathcal{P}_k\right] \leq \alpha_k^2 A.$$

We define the sequences $\phi_k, \widetilde{\psi}_k$ as follows:

$$\phi_k := \prod_{i=1}^{k-1}\frac{1}{1+\alpha_i^2 B} \text{ and } \widetilde{\psi}_k := \phi_k\psi_k.$$

Note that $0 < \lim_{k\to\infty}\phi_k := \phi_\infty < \infty$ (this can be shown by considering $\log\phi_k$ and using the condition on the sum of the squared learning rate). By substituting these sequences into the above inequality, we obtain

$$\mathbb{E}_{B_k}\left[\widetilde{\psi}_{k+1}-\widetilde{\psi}_k\middle|\mathcal{P}_k\right] \leq \alpha_k^2\phi_{k+1}A.$$

By defining $\delta_k(u) := \left(\mathbb{E}\left[u_{k+1}-u_k\right]\right)^+$, for some process u_k, we can bound the positive expected variations of $\widetilde{\psi}_k$, as follows

$$\begin{aligned}\mathbb{E}\left[\delta_k\left(\widetilde{\psi}\right)\right] =&\,\mathbb{E}\left[\left(\mathbb{E}_{B_k}\left[\widetilde{\psi}_{k+1}-\widetilde{\psi}_k\middle|\mathcal{P}_k\right]\right)^+\right]\\ \leq&\,\alpha_k^2\phi_{k+1}A.\end{aligned}$$

Due to Assumption A2, the sum of this expectation is finite. By the Quasi-Martingale Convergence Theorem, $\widetilde{\psi}_k$ converges almost surely. And, since ϕ_k converges to $\phi_\infty > 0$, ψ_k converges almost surely. Suppose $\lim_{k\to\infty}\psi_k = \psi_\infty > 0$.

If $\{w_k\}_{k=1}^\infty$ is unbounded, then for sufficiently large $k \geq \kappa$, $||w_k||^2 > D+1$ and $\psi'(||w_k||^2) \geq c_1 > 0$. Without loss of generality, we assume this instead of dealing with a subsequence. Under these conditions, the given inequality becomes equality

$$\begin{aligned}\psi_{k+1}-\psi_k =&\left(-2\alpha_k w_k^T p_k+\alpha_k^2||p_k||^2\right)\psi'(||w_k||^2)\\ &+\left(-2\alpha_k w_k^T p_k+\alpha_k^2||p_k||^2\right)^2.\end{aligned}$$

Therefore, we can express ψ_∞ as the infinite sum

$$\begin{aligned}\psi_\infty-\psi_\kappa =&\sum_{k=\kappa}^{\infty}\left[\psi_{k+1}-\psi_k\right]\\ =&\sum_{k=\kappa}^{\infty}\Big[\left(-2\alpha_k w_k^T p_k+\alpha_k^2||p_k||^2\right)\psi'(||w_k||^2)\\ &+\left(-2\alpha_k w_k^T p_k+\alpha_k^2||p_k||^2\right)^2\Big].\end{aligned}$$

The next statements hold almost surely. We have

$$\sum_{k=\kappa}^{\infty}\left(-2\alpha_k w_k^T p_k+\alpha_k^2\|p_k\|^2\right)^2 \leq \sum_{k=\kappa}^{\infty}\alpha_k^2\left(A+B\psi_k\right).$$

This can be seen by expanding the square and using Cauchy-Schwarz and Lemma 4.3. Since ψ_k converges almost surely, it is bounded above by $\psi_k \leq c_2$ almost surely. Defining $c_3 := A+Bc_2$, we have

$$\sum_{k=\kappa}^{\infty}\left(-2\alpha_k w_k^T p_k+\alpha_k^2||p_k||^2\right)^2 \leq \sum_{k=\kappa}^{\infty}\alpha_k^2 c_3.$$

Assumption A2 implies the convergence of the series on the right-hand side. Because the terms of the sum on the left are non-negative, this sum also converges almost surely by the Monotone Convergence Theorem. Similarly, the sum $\sum_{k=\kappa}^{\infty}\alpha_k^2||p_k||^2\psi'(||w_k||^2)$ converges almost surely. This uses Assumption A2, Lemma 4.3, and that ψ_k and $\psi'(||w_k||^2)$ are positive and almost surely bounded above.

Now, we subtract these convergent series from the equation for ψ_∞ to get

$$\begin{aligned}&\psi_\infty-\psi_\kappa-\sum_{k=\kappa}^{\infty}\alpha_k^2\|p_k\|^2\psi'(\|w_k\|^2)\\ &-\sum_{k=\kappa}^{\infty}\left(-2\alpha_k w_k^T p_k+\alpha_k^2\|p_k\|^2\right)^2\\ =&\sum_{k=\kappa}^{\infty}\Big[\left(-2\alpha_k w_k^T p_k+\alpha_k^2\|p_k\|^2\right)\psi'(\|w_k\|^2)\\ &+\left(-2\alpha_k w_k^T p_k+\alpha_k^2\|p_k\|^2\right)^2\Big]\\ &-\sum_{k=\kappa}^{\infty}\alpha_k^2\|p_k\|^2\psi'(\|w_k\|^2)\\ &-\sum_{k=\kappa}^{\infty}\left(-2\alpha_k w_k^T p_k+\alpha_k^2\|p_k\|^2\right)^2.\end{aligned}$$

Since the involved series almost surely converge, we can combine the terms to obtain

$$\begin{aligned}\psi_\infty - \psi_\kappa - \sum_{k=\kappa}^{\infty} \Big[& \alpha_k^2 \|p_k\|^2 \psi'(\|w_k\|^2) \\ & + \left(-2\alpha_k w_k^T p_k + \alpha_k^2 \|p_k\|^2\right)^2 \Big] \\ = \sum_{k=\kappa}^{\infty} & -2\alpha_k w_k^T p_k \psi'(\|w_k\|^2).\end{aligned}$$

Let us consider the sum on the right-hand side. By Assumption A6, $w_k^T \nabla g_k \geq c_4 > 0$ and thus

$$\sum_{k=\kappa}^{\infty} -2\alpha_k w_k^T p_k \psi'(\|w_k\|^2) \leq \sum_{k=\kappa}^{\infty} -2\alpha_k c_4 c_1.$$

Assumption A2 implies that this sequence diverges to negative infinity. This yields a contradiction, since the left-hand side must be a finite value. Therefore, $\{w_k\}_{k=1}^{\infty}$ must be bounded. □

Next, we prove that SGD converges almost surely (Theorem 4.1).

Proof. All statements here are taken almost surely. By Assumption A1, we have $f \in \mathcal{C}^5$. From linear algebra, the Hessian of $\rho(w)$ is continuous (the largest eigenvalue is a continuous function of a parametric matrix with continuous functions). This implies that $g \in C^3$, and thus, by Lemma 4.4 it is bounded on the set of all iterates. We can bound differences in the loss criteria g_k using a first-order Taylor expansion and bounding the second derivatives with K_1.

$$|g_{k+1} - g_k + \alpha_k p_k^T \nabla g_k| \leq \alpha_k^2 ||p_k||^2 K_1.$$

This can be rewritten as:

$$g_{k+1} - g_k \leq -\alpha_k p_k^T \nabla g_k + \alpha_k^2 ||p_k||^2 K_1.$$

Taking the expectation, we get

$$\begin{aligned}\mathbb{E}_{B_k}\left[g_{k+1} - g_k \middle| \mathcal{P}_k\right] \leq & -\alpha_k \, \mathbb{E}_{B_k}\left[p_k^T \nabla g_k \middle| \mathcal{P}_k\right] \\ & + \alpha_k^2 \, \mathbb{E}_{B_k}\left[||p_k||^2 \middle| \mathcal{P}_k\right] K_1.\end{aligned}$$

We decompose $p_k = \nabla g_k + (p_k - \nabla g_k)$ and bound the expectation using Lemmas 4.3 and 4.4

$$\mathbb{E}_{B_k}\left[||p_k||^2 \middle| \mathcal{P}_k\right] \leq A_2 + B_2 ||w_k||^2 \leq K_2.$$

This yields

$$\begin{aligned}\mathbb{E}_{B_k}\left[g_{k+1} - g_k \middle| \mathcal{P}_k\right] \leq & -\alpha_k ||\nabla g_k||^2 + \alpha_k^2 K_1 K_2 \\ & -\alpha_k \, \mathbb{E}_{B_k}\left[(p_k - \nabla g_k)^T \nabla g_k \middle| \mathcal{P}_k\right].\end{aligned} \tag{1}$$

Next, we apply the Cauchy-Schwarz inequality and bound the error term. From our proof to Lemma 4.2, we have

$$\left|\left| v_k^T \nabla H_k v_k - \bar{v}_k^T \nabla H_k \bar{v}_k \right|\right| \leq 2L\varepsilon_k = C_1 \varepsilon_k.$$

This implies

$$\left|\left| \mathbb{E}_{B_k}\left[p_k \middle| \mathcal{P}_k\right] - \nabla g_k \right|\right| \leq C_1 \varepsilon_k.$$

We also bound $||\nabla g_k|| \leq C_2$ using Lemma 4.4. Combining these bounds yields

$$\left|\left| \mathbb{E}_{B_k}\left[p_k \middle| \mathcal{P}_k\right] - \nabla g_k \right|\right| ||\nabla g_k|| \leq \varepsilon_k K_3. \tag{2}$$

Applying (2) to (1) gives us

$$\mathbb{E}_{B_k}\left[g_{k+1} - g_k \middle| \mathcal{P}_k\right] \leq \alpha_k^2 K_1 K_2 + \alpha_k \varepsilon_k K_3.$$

The positive expected differences are then bounded by

$$\begin{aligned}\mathbb{E}_{B_k}\left[\delta_k(h) \middle| \mathcal{P}_k\right] = & \mathbb{E}_{B_k}\left[\delta \, \mathbb{E}_{B_k}\left[g_{k+1} - g_k \middle| \mathcal{P}_k\right]\right] \\ \leq & \alpha_k^2 K_1 K_2 + \alpha_k \varepsilon_k K_3.\end{aligned}$$

By the Quasi-Martingale Convergence Theorem, g_k converges almost surely,

$$g_k \xrightarrow[k\to\infty]{\text{a.s.}} g_\infty.$$

Since g_k converges, $\sum_{k=1}^{\infty} \mathbb{E}_{B_k}\left[g_{k+1} - g_k \middle| \mathcal{P}_k\right]$ also converges. Furthermore, the series $\sum_{k=1}^{\infty} \alpha_k^2 K_1 K_2$ and $\sum_{k=1}^{\infty} \alpha_k \varepsilon_k K_3$ converge due to Assumption A2. From (1) we have

$$\sum_{k=1}^{\infty} \alpha_k ||\nabla g_k||^2 < \infty. \tag{3}$$

We define $\theta_k = ||\nabla g_k||^2$. The differences of θ_k are bounded using the Taylor expansion, similarly to the differences of g_k

$$\theta_{k+1} - \theta_k \leq -2\alpha_k p_k^T \nabla^2 g_k \nabla g_k + \alpha_k^2 ||p_k||^2 K_4,$$

for some constant K_4. Taking the expectation, we decompose p_k and bound $||p_k||^2$ similarly to (1).

$$\begin{aligned}\theta_{k+1} - \theta_k \leq & -2\alpha_k \nabla g_k^T \nabla^2 g_k \nabla g_k + \alpha_k^2 K_2 K_4 \\ & -2\alpha_k \, \mathbb{E}_{B_k}\left[(p_k - \nabla g_k)^T \nabla^2 g_k \nabla g_k \middle| \mathcal{P}_k\right]\end{aligned}$$

We also bound the second derivative by $||\nabla^2 g_k|| \leq K_5/2$ and the error term using (2), yielding

$$\begin{aligned}\mathbb{E}_{B_k}\left[\theta_{k+1} - \theta_k \middle| \mathcal{P}_k\right] \leq & \alpha_k ||\nabla g_k||^2 K_5 + \alpha_k^2 K_2 K_4 \\ & + \alpha_k \varepsilon_k K_3 K_5.\end{aligned}$$

The positive expectations are bounded,

$$\begin{aligned}\mathbb{E}_{B_k}\left[\delta_k(\theta) \middle| \mathcal{P}_k\right] = & \mathbb{E}_{B_k}\left[\delta \, \mathbb{E}_{B_k}\left[\theta_{k+1} - \theta_k \middle| \mathcal{P}_k\right]\right] \\ \leq & \alpha_k ||\nabla g_k||^2 K_5 + \alpha_k^2 K_2 K_4 \\ & + \alpha_k \varepsilon_k K_3 K_5.\end{aligned}$$

Since the terms on the right-hand side are sums of convergent infinite sequences (due to Assumption A2 and (3)), by the Quasi-Martingale Convergence Theorem, θ_k converges almost surely. Suppose $||\nabla g_k||$ converges to a positive value $C_3 > 0$. Then for sufficiently large $k \geq \kappa$ there exists a positive constant $0 < C_4 < C_3$ such that $||\nabla g_k|| \geq C_4$. Thus, $\sum\limits_{k=\kappa}^{\infty} \alpha_k ||\nabla g_k||^2 \geq C_4^2 \sum\limits_{k=\kappa}^{\infty} \alpha_k$. By Assumption A2, this diverges, contradicting (3). Therefore, the limit must be zero

$$\theta_k \xrightarrow[k\to\infty]{\text{a.s.}} 0 \text{ and } \nabla g_k \xrightarrow[k\to\infty]{\text{a.s.}} 0.$$

□

Appendix C
Additional Experiment Details

The code is available in the anonymized online repository. The algorithm is written in Python, using PyTorch [3] and TorchVision [4]. Forest cover-type and USPS experiments are run on a 3.1 GHz Dual-Core Intel Core i5 processor with 16 GB 2133 MHz LPDDR3 memory. Chest X-ray experiments are run on an Intel Xeon CPU E5-2650 v4 @ 2.20GHz with an NVIDIA Tesla K40c GPU.

A. Data Sets

The forest cover-type data [5] uses cartographic data to predict the tree species (as determined by the United States Forest Service) of a 30 x 30-meter cell. This cartographic data includes elevation, aspect, slope, distance to surface water features, distance to roadways, hill-shade index at three times of day, distance to wildfire ignition points, wilderness area designation, and soil type. Seven major tree species are included: spruce/fir, lodgepole pine, Ponderosa pine, cottonwood/willow, aspen, Douglas-fir, and krummholz. In total, 581,012 samples are included, which we split 64%/16%/20% into train/validation/test data sets.

The USPS digits data [6] includes 16 x 16 pixel greyscale images from scanned envelopes to identify which digit 0-9 each image corresponds to. This data set is already split into 7,291 training and 2,007 test images. We take 1/7 of the training set as validation.

The chest X-ray data [7] contains 1024 x 1024 pixel color images of patients' chest X-rays, to identify which of fourteen lung diseases each patient has. Note that this is a multi-label problem; patients can have none, one, or multiple of these conditions. A total of 112,120 patients' images are included, taken between the years 1992 and 2015, which we split 70%/10%/20% into train/validation/test data sets.

B. Generalization Tests

For forest cover-type data, we weight the test subjects to shift the mean of a feature or multiple features. Since we normalize the data, the weight of each test subject is determined by the ratio of the normal probability distribution function value with and without the shift. This shift adds a slight bias to the test set that is not in the original data set. We first use this shift method to increase the mean of each feature value by 0.1, compare the accuracy of our trained models, and find that certain features are problematic. Upon further examination, this is because these features are binary factors with rare classes (so our weighting of subjects emphasizes a few of them). Then, we shift each feature (except the problematic features) by a random normal amount (with mean 0 and standard deviation 0.05), compare the accuracy of our trained models, and repeat it one thousand total times.

For USPS handwritten digits data, we augment the test set: Augmented Test 1 uses random crops (with padding) of up to one pixel and random rotations of up to 15°; Augmented Test 2 uses crops of up to two pixels and rotations of up to 30°. Note that we do not augment our training set while learning our models, as a similar augmentation would yield comparable training and test sets.

For the Conditional GAN examples, we modify the implementation of Linder-Norén [8] and generate 10,000 images from the trained generator model. We train the GAN model with a batch size of 64, cosine annealing learning rate (initially 10^{-4}), $\beta_1 = 0.5$, and $\beta_2 = 0.999$. We randomly smooth the labels to be uniform between 0.0 and 0.3 for generated samples and 0.7 and 1.0 for true samples. We also swap the labels on 1% of batches, chosen at random. We also modify the implementation of Chhabra [9] and generate 10,000 images from the trained generator model.

For the chest X-ray data, we compare performance on two similar data sets, CheXpert [10] and MIMIC-CXR [11]. For this comparison, we only consider the six conditions common to the three data sets: atelectasis, cardiomegaly, consolidation, edema, pneumonia, and pneumothorax. Additionally, we ignore any uncertain labels in the CheXpert and MIMIC-CXR classes. We keep the assigned training and validation data sets separate for each data set, as there appear to be differences in labeling. Particularly, the CheXpert validation set is fully labeled, while the training set contains uncertain and missing labels. While these are labeled "training" and "validation" sets, we solely use them as test sets. CheXpert contains 234 validation and 223,415 training images from Stanford Hospital, taken between 2002 and 2017. MIMIC-CXR contains 2,732 validation and 369,188 training images from Beth Israel Deaconess Medical Center, taken between the years 2011 and 2016.

We measure the spectral radius ρ of each model on the full training set. We use $\varepsilon = 10^{-3}$ and a maximum of 1,000 power iterations, except for the chest X-ray models, where we use $\varepsilon = 0.1$ and a maximum of 100 power iterations. These values allow the algorithm to find an accurate eigenvalue within a reasonable run time.

C. Implementation

For forest cover-type models, we train using stochastic gradient descent, with learning rate $\frac{0.5}{\text{epoch \#}}$, batch size of 128, and a maximum of 100 epochs. For models with batch sizes 32 and 64, we use a $\frac{0.1}{\text{epoch \#}}$ learning rate instead. The network uses 3 hidden layers with 20 hidden nodes in each layer. The learning rate and maximum number of epochs allow our algorithm to converge to an accurate model. We experiment with other feed-forward networks, learning rates, and optimizers but find that this structure works best. Our experiments with batch size are discussed in Section 5.3.

For USPS models, we use the Adam optimizer with a learning rate of 10^{-3}, a batch size of 128, and a maximum of 100 epochs. The network takes an input image and processes it through the following layers in order: convolution to 8 channels, pool, convolution to 16 channels, pool, convolution to 32 channels, pool, fully connected to 64 nodes, fully connected to 10 nodes, softmax. All convolutions are of kernel size 3, stride 1, and padding 1; all pools are max pooling with kernel size 2 and stride 2. ReLUs are used to connect the various layers. The MNIST images are re-sized to be 16 x 16 to match the model's input size. We similarly experiment with feed-forward and other convolutional networks, learning rates, and optimizers, but find that this structure performs best.

For the chest X-ray models, we resize the images to 256 x 256 and then crop them to 224 x 224. We use the Adam optimizer with a learning rate of 10^{-5}, batch size of 4, a maximum of 100 power iterations, random initialization of each power iteration, and gradient clipping (at a magnitude of 100). The crops and small batch size are necessary for the GPUs on our server to not run out of memory (12 GB). Using a CheXNet model [12] (trained using their methodology) as the initialization, we train for an additional epoch, except for the Asymmetric Valley model (which we address later). We also try 5 epochs; however, the first epoch has the highest validation mean AUC in each case.

The entropy-SGD models are trained with a learning rate of 0.1 (except on chest X-ray data, where 0.001 is used), a momentum of 0.9, and no dampening or weight decay. The K-FAC models are trained using the implementation of Wang [13], with a learning rate of 0.001 (except on chest X-ray data, where 10^{-7} is used) and a momentum of 0.9.

The Asymmetric Valley models are trained with an initial learning rate of 0.5 for 250 epochs (iterations 161-200 utilizing SWA). For chest X-ray data, we start at the SWA step, using CheXNet initialization [12].

The forest cover-type LOBPCG model is trained with regularization parameters $\mu = .0028$ and $K = 1$, update frequency $b = 4$ and learning rate $\tilde{\alpha}(j) = \exp(-4j - 2)$. The USPS LOBPCG model is trained with regularization parameters $\mu = .005$ and $K = 0$, update frequency $b = 4$ and learning rate $\tilde{\alpha}(j) = \exp(-4j)$.

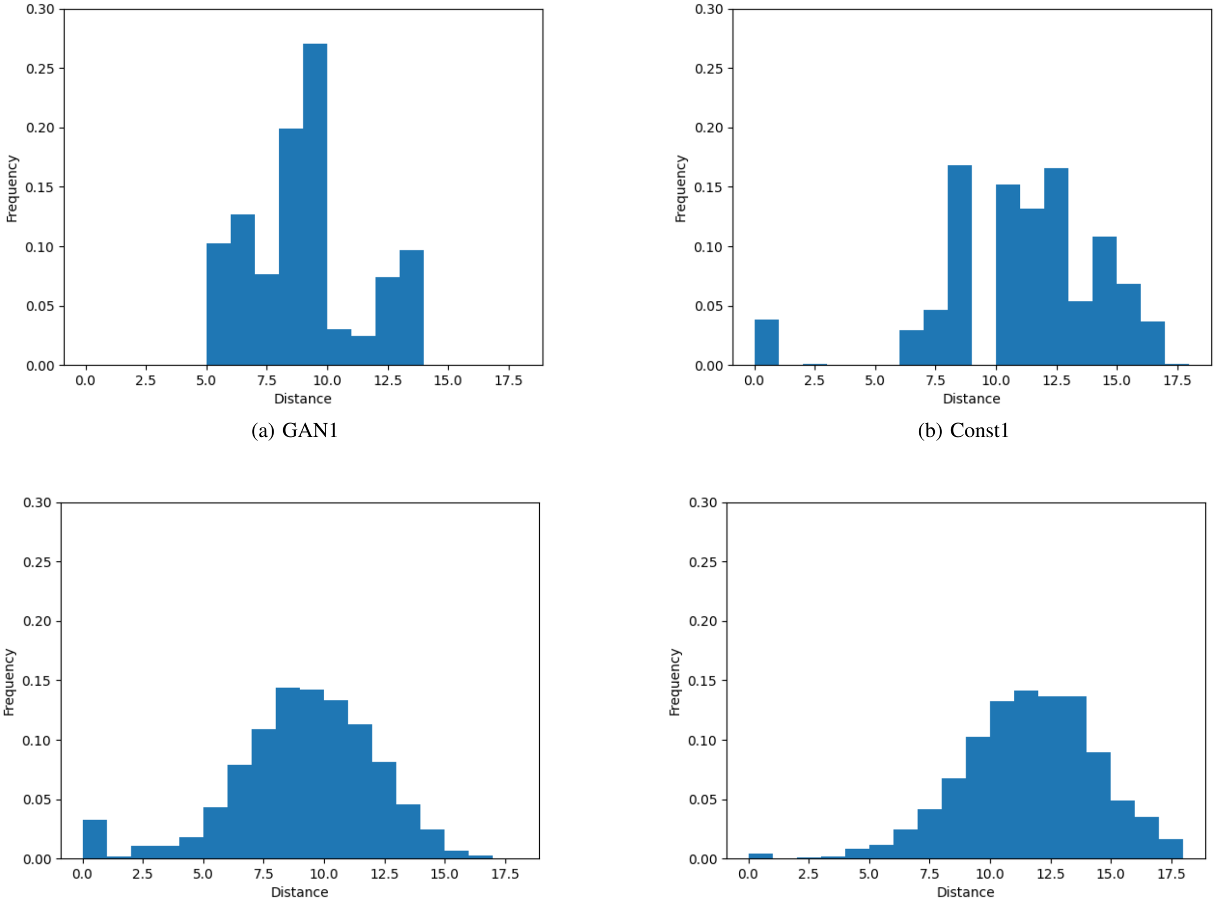
APPENDIX D
CONSTRUCTED DATA SETS

During analysis of the performance of GAN1, we compute the minimum distance (L_2-norm) between each image in the GAN1 data set and the images in the USPS test data. We notice that the GAN1 images are distributed differently, relative to the USPS test data, compared to the other data sets. In particular, Figure 6 shows that the augmented data sets have a bell-curve distribution of such distances, while GAN1 has an abnormal distribution. We construct a data set, Const1, from the augmented test data sets using the following procedure.

1) Split the data with respect to distances into integer bins [0,1), [1,2), $\cdots$, [17,18).
2) Uniformly at random, select 5 bins to draw zero images from.
3) For the remaining bins, select one of the two augmented test data sets at uniform random. Add all images from the selected data set in the bin's distance range to the constructed data set.

This procedure creates a constructed data set intended to emulate the abnormal distribution of the GAN1 data.

We repeat this process with maximum cosine similarity between images and observe similar distributional abnormalities in the GAN1 data (Figure 7). We construct Const2 from the augmented data set using a similar procedure, but with bins [0.5, 0.525), [0.525, 0.55), $\cdots$, [0.975, 1.0).



(a) GAN1

(b) Const1

(c) AT1

(d) AT2

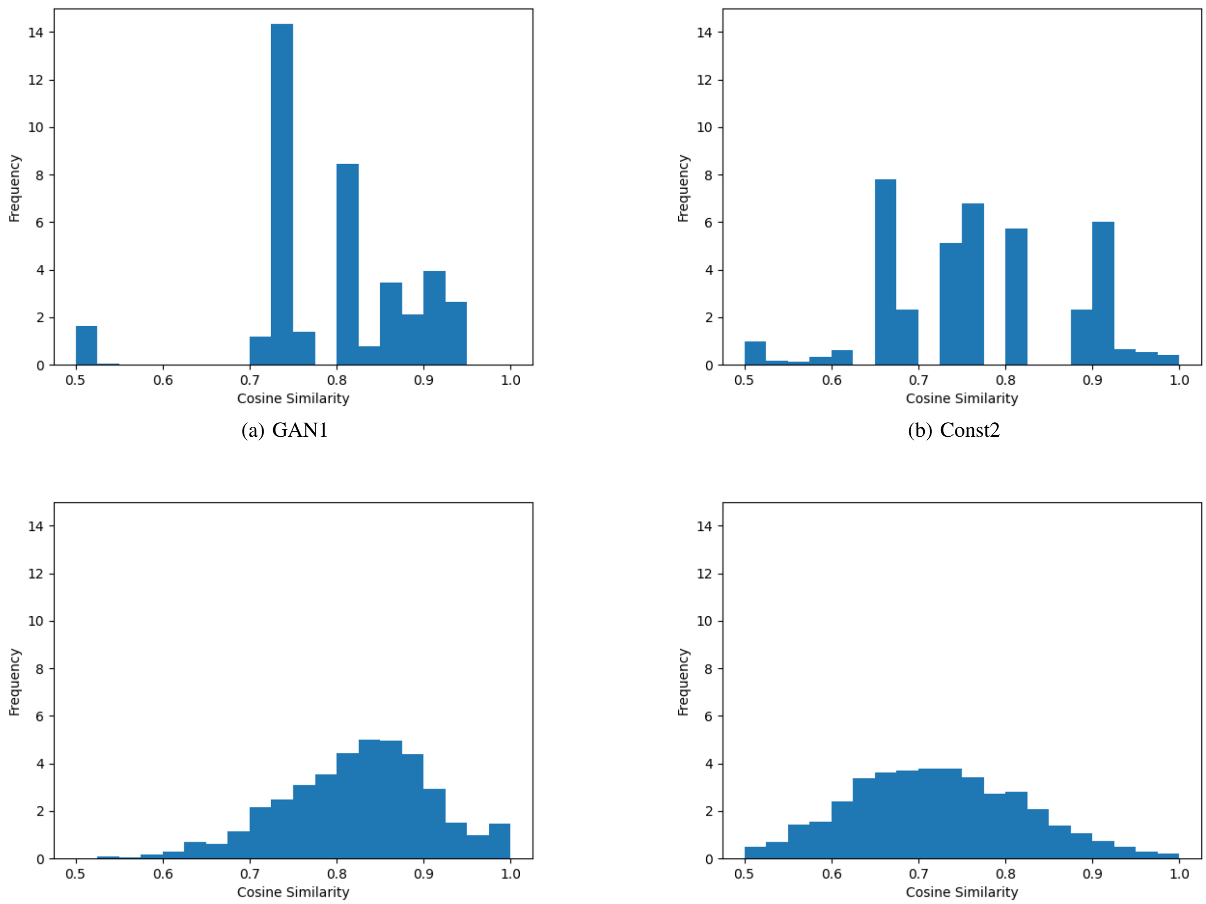
Fig. 6. Histograms of Euclidean distance between data sets and USPS test data.

Appendix E
Grad-CAM

We use Gildenblat's Grad-CAM implementation [14] to highlight which areas of the chest X-rays are important to our best regularized model ($\mu = 10^{-4}$ and $\alpha = 10^{-6}$) and the predictions of the baseline models. We compute the Jaccard index of the top 10% of pixels in each Grad-CAM image to compare which regions are important to each model.

Tables VI and VII show that the two models with the lowest spectral radius ρ, our regularized model and entropy-SGD, have the highest overlap in explanations. These models highlight similar areas of the chest X-rays as important in making predictions. The Jaccard scores of their overlap are over .5 on the CheXpert and MIMIC-CXR validation data, the highest of any pair of models. Since these models also perform best on these transfer learning data, evidence suggests that models with low spectral radius generalize better in both their explanations and predictions. The three models with higher spectral radii have a larger dip in performance and less overlap in their explanations.

In contrast, the three models with higher spectral radius, unregularized, K-FAC, and asymmetric valley, have a higher performance drop on the transfer learning data sets and have less overlap in their explanations. These models have mean Jaccard scores of .202-.306 on the CheXpert Validation data, lower than the spectral radius regularization and EntropySGD scores of .342 and .351. The high spectral radius models have scores of .199-.249 on MIMIC-CXR Validation data, while the low spectral radius models have scores of .330 and .316.



(a) GAN1 (b) Const2

(c) AT1 (d) AT2

Fig. 7. Histograms of cosine similarity between data sets and USPS test data.

TABLE VI

THERE IS MORE OVERLAP IN THE EXPLANATIONS FROM THE TWO MODELS WITH LOW SPECTRAL RADIUS ρ (OUR SPECTRAL RADIUS REGULARIZATION AND ENTROPY-SGD) ON THE CHEXPERT VALIDATION DATA SET.

Model	ρ	Jaccard Score of Overlap on CheXpert Validation Data						PerfDrop
		SpecRad	EntropySGD	UnReg	KFAC	AsymValley	Mean	
SpecRad	38.92	1.000	0.508	0.192	0.294	0.374	0.342	-4.15%
EntropySGD	41.11	0.508	1.000	0.181	0.280	0.435	0.351	-3.09%
UnReg	68.29	0.192	0.181	1.000	0.283	0.152	0.202	-10.30%
KFAC	84.65	0.294	0.280	0.283	1.000	0.261	0.280	-13.16%
AsymValley	1198.69	0.374	0.435	0.152	0.261	1.000	0.306	-10.06%

TABLE VII

THERE IS MORE OVERLAP IN THE EXPLANATIONS FROM THE TWO MODELS WITH LOW SPECTRAL RADIUS ρ ON THE MIMIC-CXR VALIDATION DATA SET.

Model	ρ	Jaccard Score of Overlap on MIMIC-CXR Validation Data						PerfDrop
		SpecRad	EntropySGD	UnReg	KFAC	AsymValley	Mean	
SpecRad	38.92	1.000	0.504	0.216	0.285	0.313	0.330	-8.10%
EntropySGD	41.11	0.504	1.000	0.172	0.229	0.358	0.316	-9.21%
UnReg	68.29	0.216	0.172	1.000	0.272	0.134	0.199	-10.80%
KFAC	84.65	0.285	0.229	0.272	1.000	0.192	0.245	-10.71%
AsymValley	1198.69	0.313	0.358	0.134	0.192	1.000	0.249	-10.84%

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